CHAPTER – 7 RESPIRATION IN PLANTS

📝 Part 1: In-Chapter Pointwise Notes

⚡ 7.1 What is Respiration?

- All living cells in plants and animals require energy for various activities (building proteins, making starch from glucose, absorbing minerals, and laying down cell walls).
- This energy is made available by the **breakdown of glucose utilizing oxygen**, represented by the following overall aerobic respiration equation:

$$C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{enzyme}} 6CO_2 + 6H_2O + ATP + \text{Heat energy}$$

.

- **Key Concept:** The release of energy occurs specifically because **bonds break** during this chemical breakdown.
- **Three Important Characteristics of Respiration:**
    1. The breakdown of glucose is not a single step but occurs in a **series of chemical steps** divided into two major phases:
        - **Glycolysis:** Breakdown of glucose into pyruvate ($C_3H_4O_3$). This occurs in the cytoplasm and yields **2 ATP**.
        - **Krebs Cycle:** Breakdown of pyruvate into CO_2 + H_2O + ATP. This occurs in the mitochondria and yields **36 ATP**.
    2. Each breakdown step is controlled by a **particular enzyme**.
    3. The energy liberated is not all heat; a large part is converted into chemical energy in the form of **ATP** (adenosine triphosphate).
- **The ATP Cycle:** When ATP is used for bodily work, it converts to ADP (adenosine diphosphate). When more energy is available from glucose breakdown, ADP is reconverted back to ATP.
- One molecule of glucose on complete oxidation yields a total of **38 molecules of ATP**.

🔄 ATP & Metabolic Activities

- ATP is the immediate source of energy available for all activities inside a living cell, earning it the title **"energy currency of the cell"**.
- *Note on Energy Storage:* Plants store energy primarily as **starch**, whereas animals store it as **glycogen**.
- Living organisms show two types of metabolic activities:
    1. **Anabolic:** Constructive or biosynthetic processes; it **consumes energy** ➡️ *(Example: Converting Glucose into stored Glycogen).*
    2. **Catabolic:** Destructive or breaking down processes; it **gives out energy** for use by the organism.
- **Definition:** Respiration is a **catabolic process** of releasing energy from glucose for carrying out life processes.

🔥 7.2 Respiration vs. Burning (Combustion)

Parameter 📏	**Respiration** 🧬	**Burning / Combustion** 🔥
Process Type	Cellular process.	Non-cellular process.
Temperature	Occurs at body temperature.	Occurs at high temperature (at ignition point).
Pacing	Occurs in a series of chemical steps.	Occurs in a single step.
Mechanism	Carried out by enzymes (Biochemical process).	Carried out by heat (Physico-chemical process).
Energy Output	Energy liberated as ATP and some heat.	All energy is liberated as heat and light.

Parameter 📏	Respiration 🧬	Burning / Combustion 🔥
Light	No light energy is produced.	Light energy is produced.

🌿 7.3 The Entire Plant Respires

- **Key Assertion/Reason Fact:** Every single part of a plant—the leaves, stem, roots, and even the deepest cell placed in any region—respires.
- **Three Inlets for Oxygen:**
    1. **Stomata** in leaves.
    2. **Lenticels** in the stem *(defined as: permanent openings in permanent stems)*.
    3. **General surface** of the roots.
- **Soil Aeration:** Ploughing or tilling creates tiny air spaces around soil particles, providing an oxygen source for roots. Conversely, water-logged and compact soil lacks air spaces, severely affecting root respiration.
- **Timing of Respiration:** Respiration occurs **both day and night** continuously.
    - **Daytime:** Leaves produce O_2 through photosynthesis; some is used for respiration, and the rest diffuses out. The CO_2 produced by respiration serves as raw material for photosynthesis.
    - **Nighttime:** Leaves directly obtain oxygen from the atmosphere and give out carbon dioxide.

🫁 7.4 Two Kinds of Respiration - Aerobic and Anaerobic

- **A. Aerobic (Oxybiotic) Respiration:**
    - Normally, free oxygen is used resulting in the **complete combustion/oxidation of glucose** into carbon dioxide and water.
    - Equation:

$$C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{enzyme}} 6CO_2 + 6H_2O + 38ATP$$

.

- **B. Anaerobic (Anoxybiotic) Respiration:**
  - Proceeds **without oxygen**.
  - Glucose is incompletely broken down into **ethanol (ethyl alcohol)** and carbon dioxide, releasing a small quantity of energy (2 ATP).
  - Equation:

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + 2ATP$$

.

  - **Mechanism Location:** This specific breakdown (Glucose → Pyruvate → Ethanol + CO_2) happens entirely within the **cytoplasm**.
  - *Animal Contrast:* In animals, anaerobic respiration follows a different path: Glucose → Pyruvate → **Lactic acid**.
  - **Occurrence:** Plant parts (fruits/seeds) may respire anaerobically **temporarily** under situations like **water-logging**, but it cannot continue for long before the plant dies.
  - However, microscopic organisms like certain bacteria and fungi (e.g., Yeast) normally respire *only* anaerobically throughout their lives.

Parameter 📏	**Aerobic Respiration** 🌬️	**Anaerobic Respiration** 🚫🌬️
Oxygen	Proceeds in the presence of oxygen.	Proceeds without using oxygen.
Breakdown	Complete breakdown of glucose.	Incomplete breakdown of glucose.

Parameter 📏	Aerobic Respiration 🫁	Anaerobic Respiration 🚫🫁
End-Products	Carbon dioxide and water.	Ethyl alcohol and carbon dioxide.
Energy Yield	Liberated in large quantity (**38 ATP**).	Liberated in small quantity (**2 ATP**).
Duration	Occurs normally throughout life.	Occurs temporarily for short periods (except in some bacteria/fungi).

🧪 7.5 Experiments on Respiration in Plants

- **Experiment 1: To prove that oxygen is used up in respiration**
  - **Setup:**
    - Flask A contains living, germinating bean seeds.
    - Flask B (the Control) contains dead (boiled) seeds washed with an antiseptic (like carbolic acid) to avoid bacterial decay.
    - Wet cotton is used to provide moisture to both.
    - A small tube containing **soda lime** (a mixture of sodium hydroxide and calcium oxide) is suspended in each flask for the express purpose of **absorbing any carbon dioxide** released.
  - **Observation:** After a few days, the delivery tube connected to Flask A shows a significant rise in water level.
  - **Reasoning:** The germinating seeds use up the oxygen, creating a **vacuum** inside Flask A. Because the air moved/shrank, water is carried up the delivery tube to fill the space.

- **Confirmation Check:** When a burning paper is introduced into Flask A, the flame is immediately put off (proving oxygen is absent). In Flask B, it continues to burn for a short while.
- *Study Prompt:* (Note the text asks to solve why there might be a *slight* increase in Flask B's delivery tube—this typically happens due to slight temperature drops or minor atmospheric pressure changes in the room affecting the sealed flask!)

📝 Part 1: In-Chapter Point-Wise Notes (Continued)

🧪 Experimental and Control Set-Ups

- A proper experiment must always have two components: an **experimental set-up** and a **control set-up**.
- Both set-ups are absolutely identical, except **the condition under study is missing** in the control set-up.

🌱 7.5 Experiments on Respiration (Continued)

- **Experiment 2: To prove that CO_2 is produced during respiration in seeds**
  - Two flasks (A and B) are set up with wet cotton wool. Flask A contains soaked, germinating seeds (pea/bean) and Flask B contains boiled (dead) seeds.
  - Vaseline is applied to the corks to seal the flasks tight and make them air-tight.
  - A little antiseptic (such as carbolic acid) is added to Flask B to prevent bacterial growth on the dead seeds, which would otherwise respire, release carbon dioxide, and ruin the control.
  - After a few days, the gases are tested by removing the cork and tilting the flask over a test tube of limewater.
  - Because the expected gas (CO_2) is **heavier than air**, it easily flows down into the test tube.
  - **Result:** Gas from Flask A turns limewater milky, proving CO_2 is produced by germinating seeds. Flask B shows no effect.

- **Alternative Method for Experiment 2 (Four-Flask Setup):**
    - An air suction pump is attached to the end to create a **partial vacuum**, pulling and maintaining a one-way flow of air through all the flasks.
    - **Flask A (KOH solution):** Dissolves and removes any atmospheric CO_2 present in the incoming air.
    - **Flask B (Limewater):** Stays clear (not milky), confirming the air entering the next flask is completely CO_2-free.
    - **Flask C (Germinating seeds):** The live seeds respire and add CO_2 to the air.
    - **Flask D (Limewater):** Turns milky, proving conclusively that the only source of the CO_2 was the germinating seeds in Flask C.
- **Experiment 3: To prove CO_2 is produced by GREEN PLANTS during respiration**
    - A potted plant (like Geranium) is placed under a bell jar with all connections vaselined to make them air-tight.
    - The incoming air is completely stripped of CO_2 by passing it through soda lime and a limewater flask (Flask A).
    - The entire bell jar must be completely covered by a **black cloth** to keep it in the dark and **prevent photosynthesis**. If left in the light, the plant would immediately use up the CO_2 liberated from respiration to synthesize starch.
    - **Result:** The outgoing air turns the limewater in Flask B milky, proving the plant respired and produced CO_2.
- **Experiment 4: To show that heat is evolved during respiration**
    - Two thermoflasks are used. Flask A contains live germinating seeds, and Flask B contains boiled seeds.
    - The boiled seeds are washed with dilute **formalin or carbolic acid** to prevent bacterial decay, which could falsely generate heat.

- **Result:** A thermometer placed in Flask A shows a higher reading, indicating that germinating (respiring) seeds give out heat.

- **Experiment 5: To demonstrate anaerobic respiration**
  - 8 to 10 soaked and **peeled** pea seeds are pushed into a test tube filled completely with, and inverted in, a beaker of mercury.
  - The seed coats (testa) are peeled off before introduction to remove the physical barrier and **hasten the outward diffusion of** CO_2.
  - Water is not used in this experiment because seeds could dissolve and extract oxygen from the H_2O, and water also absorbs CO_2. Mercury prevents both issues.
  - **Result:** The level of mercury falls as the liberated gas collects at the top.
  - To test the gas, a stick of **potassium hydroxide (KOH)** is introduced. It floats up through the mercury and immediately absorbs the CO_2 gas, causing the level of mercury to rise again.

⚖️ 7.6 Respiration Contrasted with Photosynthesis

- In many respects, respiration and photosynthesis in plants are **distinctly opposite processes**. The requirements of one are the products of the other, making them complementary.
- **Timing dynamic:** During the day, there is significantly more photosynthesis occurring than respiration.
- **Photosynthesis Equation:** $CO_2 + H_2O \xrightarrow{\text{light/chloro}} \text{Glucose} + O_2$ gas.

Parameter 📏	**Photosynthesis** 🌞	**Respiration** 🧬
Occurrence	Occurs only in the presence of chlorophyll.	Occurs in all living cells.

Parameter 📏	Photosynthesis 🌞	Respiration 🧬
Timing	Occurs only in the presence of light.	Occurs at all times.
Inputs	Uses carbon dioxide and water.	Uses oxygen and glucose.
Outputs	Oxygen is released as an end product.	Carbon dioxide and H_2O are released as end products.
Energy	Light energy is converted into stored chemical energy.	Chemical energy is partly converted into heat and partly into useful energy.
Weight	Results in **gain in weight**.	Results in **loss in weight**.
Metabolic Type	Food (glucose) is manufactured (Constructive / anabolic process).	Food (glucose) is broken down / oxidised (Destructive / catabolic process).

🐾 7.7 Respiration in Plants Compared with Respiration in Animals

The basic aspects of respiration are the same in all organisms, but there are three key differences between plants and animals:

1. **Gaseous Transport:** In plants, there is **no gaseous transport**; respiratory gases simply diffuse in and out via cell-to-cell diffusion. In animals, the blood physically transports respiratory gases.
2. **Anaerobic End Products:** One of the end products of anaerobic respiration in plants is **ethanol/ethyl alcohol**, while in animals it is **lactic acid**.

3. **Heat Production:** In plants, the process produces **little heat** as compared to animals.

🌳 **Myth-Busting: Sleeping Under a Tree at Night**

- **The Myth:** People often say you shouldn't sleep under a tree at night because of the carbon dioxide it releases.
- **The Reality:** The CO_2 released by trees during respiration at night is actually **too little to cause any harm** (we often have much more CO_2 trapped in our own bedrooms!).
- **The Real Dangers:** The actual risks of sleeping under a tree at night are the **droppings of perching birds** and the presence of **night-active insects, spiders, snakes, etc.**
- **Daytime Benefits:** Sleeping under a tree during a hot mid-day is definitely good because you get both oxygen (due to photosynthesis) and coolness (due to transpiration).

🏆 **Part 2: End-of-Chapter Master Lists (Respiration in Plants)** 🌿🍽️

📚 **1. Combined Terms List (Glossary)** 🧠

Biological Term 🔑	**Exam-Ready Definition** 📝
Respiration	A **catabolic process** in which living cells oxidise organic substances (glucose) to release carbon dioxide, water vapour, and energy.
Glycolysis	The first phase of respiration occurring in the cytoplasm where **glucose is broken down into pyruvate**, yielding 2 ATP.
Krebs Cycle	The second phase of respiration occurring in the mitochondria where **pyruvate is broken down into** CO_2 **and** H_2O, yielding 36 ATP.

Biological Term 🔑	Exam-Ready Definition 📝
ATP	Adenosine triphosphate; the chemical compound acting as the **immediate "energy currency"** of the cell.
ADP	Adenosine diphosphate; the compound formed **when energy from ATP is used**, which can be reconverted to ATP.
Anabolic Process	Constructive or biosynthetic metabolic activities that **consume energy** (e.g., making starch from glucose).
Catabolic Process	Destructive or breaking down metabolic activities that **give out energy** (e.g., respiration).
Combustion (Burning)	A fast, non-cellular, physico-chemical process occurring at high temperatures in a **single step**, liberating energy as heat and light.
Aerobic (Oxybiotic) Respiration	Respiration that proceeds in the **presence of oxygen**, resulting in the complete breakdown of glucose (yielding 38 ATP).
Anaerobic (Anoxybiotic) Respiration	Respiration that proceeds **without using oxygen**, resulting in the incomplete breakdown of glucose (yielding 2 ATP, ethanol, and CO_2).
Lenticels	**Permanent openings** found in older, permanent stems that serve as inlets for oxygen.

Biological Term 🔑	Exam-Ready Definition 📝
Stomata	Tiny pores found primarily on **leaves** that allow for the exchange of respiratory gases.

🌟 2. Master Example List 🌿

- **Metabolic Processes:**
  - *Anabolic (Constructive):* Building proteins from amino acids, making starch from glucose.
  - *Catabolic (Destructive):* Respiration.
- **Inlets for Oxygen in Plants:**
  - *Leaves:* Stomata.
  - *Stems:* Lenticels.
  - *Roots:* General surface.
- **Anaerobic Respiration Organisms/Tissues:**
  - *Microorganisms (Normal/Lifelong):* Yeast, certain bacteria, and fungi.
  - *Plant Parts (Temporary):* Fruits and seeds (especially during water-logging).
- **End Products of Anaerobic Respiration:**
  - *In Plants:* Ethanol (ethyl alcohol) + CO_2.
  - *In Animals:* Lactic acid.
- **Energy Storage Forms:**
  - *In Plants:* Starch.
  - *In Animals:* Glycogen.
- **Dangers of Sleeping Under Trees at Night:**
  - Bird droppings, night-active insects, spiders, snakes (NOT carbon dioxide).

⚙️ 3. Master Function & Location Matrix 📍🛠️

Biological Part / Setup Chemical 🧬	Exact Location / Placement 📍	Primary Function / Action ⚙️
Glycolysis (Phase 1)	Cytoplasm.	Breaks down glucose into pyruvate (produces 2 ATP).
Krebs Cycle (Phase 2)	Mitochondria.	Breaks down pyruvate into CO_2 and H_2O (produces 36 ATP).
ATP (Energy Currency)	Inside all living cells.	Provides immediate chemical energy for cellular work.
Carbolic Acid / Formalin	Added to the flask with boiled/dead seeds.	Acts as an antiseptic to prevent bacterial decay (which would falsely release CO_2 or heat).
Vaseline	Applied to corks and glass rims in experiments.	Seals connections to make the apparatus completely air-tight.
Soda Lime	Placed in incoming air tubes / suspended in flasks.	A mixture of NaOH and CaO that absorbs any atmospheric CO_2.
Limewater	Inside test tubes or flasks (Flask B/D).	Turns milky in the presence of CO_2; used to test for its presence.

Biological Part / Setup Chemical 📌	Exact Location / Placement 📍	Primary Function / Action ⚙️
Potassium Hydroxide (KOH)	Solution in flasks or solid stick in mercury tube.	Strongly absorbs carbon dioxide gas.
Black Cloth	Covering the bell jar in the plant respiration experiment.	Prevents light from entering to stop photosynthesis (so the plant doesn't consume the respired CO_2).
Mercury	Filling the beaker and test tube in the anaerobic experiment.	Acts as a liquid seal because it does not dissolve seeds, provide oxygen, or absorb CO_2 (unlike water).
Air Suction Pump	Attached to the end of the four-flask setup.	Creates a partial vacuum to pull and maintain a one-way flow of air.

Page 69

Question: 1. Write the important aspects about the overall chemical equation of respiration, pertaining to the following: (i) number of steps for glucose to break down into CO_2 and H_2O. (ii) involvement of enzymes. (iii) forms of energy liberated.

Answer:

- (i) The breakdown of glucose occurs in a series of chemical steps, rather than a single step.
- (ii) Each step in the breakdown process is carried out by a particular enzyme.

- (iii) Energy is liberated as chemical energy in the form of ATP alongside some heat.

Question: 2. List three ways in which respiration is different from burning.

Answer:

- Respiration is a cellular, biochemical process carried out by enzymes, whereas burning is a non-cellular, physico-chemical process carried out by heat.
- Respiration occurs at body temperature in a series of chemical steps, while burning occurs at a high ignition temperature in a single step.
- Respiration liberates energy as ATP and heat with no light produced, whereas burning liberates all energy as heat and light.

Question: 3. Name the three inlets of oxygen for respiration in plants.

Answer: The three inlets of oxygen are stomata in the leaves, lenticels in the stem, and the general surface of the roots.

Page 70

Question: 1. Give the overall chemical equation for the two kinds of respiration in plants : (i) Aerobic (ii) Anaerobic

Answer:

- **(i) Aerobic respiration:** $C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{enzyme}} 6CO_2 + 6H_2O + 38ATP.$
- **(ii) Anaerobic respiration:** $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + 2ATP.$

Question: 2. Mention any one difference other than those reflected in the two chemical equations given above.

Answer: Aerobic respiration occurs normally throughout the life of the plant, whereas anaerobic respiration in plants occurs only temporarily for short periods.

Question: 3. Certain organisms respire only anaerobically throughout their lives. Name any two such organisms.

Answer: Certain bacteria and fungi, such as yeast, respire only anaerobically.

Page 73

Question: 1. In experiments on respiration, the seeds taken as control are boiled. Why?

Answer: Seeds are boiled to kill them, ensuring they do not respire. This creates a proper control setup where the specific condition under study (respiration) is missing.

Question: 2. What is the use of the following in the experiments on respiration ? (i) Soda lime (ii) Limewater

Answer:

- **(i)** Soda lime is used to absorb any carbon dioxide present in the air.
- **(ii)** Limewater is used to test for the presence of carbon dioxide, as it turns milky when exposed to the gas.

Question: 3. Suppose we conduct an experiment to demonstrate respiration in a green plant. What special precaution is required to be taken in it?

Answer: The experiment must be carried out in the dark, or the plant must be completely covered by a black cloth. This prevents photosynthesis from occurring, which would otherwise use up the carbon dioxide liberated during respiration.

Question: 4. In experiment no. 5, we are using mercury in the test tube and the beaker. Why can't we use water instead?

Answer: Water cannot be used because the seeds might dissolve in it or extract oxygen from it, which would ruin the anaerobic conditions. Furthermore, water absorbs carbon dioxide gas, which would prevent the liberated gas from collecting and being accurately observed.

Page 74

Question: 1. Fill in the blanks : In anaerobic respiration in plants, the end product is ..... and ..... instead of ..... in animals.

Answer: In anaerobic respiration in plants, the end product is **ethyl alcohol (ethanol)** and **carbon dioxide** instead of **lactic acid** in animals.

Question: 2. The table below is intended to give certain differences between photosynthesis and respiration, but a few points are given wrongly. Strike out the wrong ones and write the correct words instead.

Answer:

- **Under Photosynthesis:** Strike out "Results in loss in weight" and write **Results in gain in weight**. Strike out "Sucrose is the product" and write **Glucose is the product**. Strike out "Uses water and oxygen" and write **Uses carbon dioxide and water**.
- **Under Respiration:** Strike out "Results in gain in weight" and write **Results in loss in weight**. Strike out "Sucrose is broken down" and write **Glucose is broken down**.

Review Questions: A. MULTIPLE CHOICE TYPE

(i) Glycolysis is a process

Answer: in which glucose is broken down into pyruvate.

(ii) In plants, a common function is performed by?

Answer: Stomata and lenticels.

(iii) Anaerobic respiration normally occurs in

Answer: Baker's yeast.

(iv) On oxidation of 1 molecule of glucose, ... ATP is produced through aerobic respiration.

Answer: 38.

(v) Respiration in plants occurs :

Answer: In all the parts.

(vi) Which of the following is produced during anaerobic respiration in plants?

Answer: Ethyl alcohol.

(vii) Glycolysis takes place in :

Answer: Cytoplasm.

(viii) Lime water turns milky upon absorption of carbon dioxide gas due to the formation of :

Answer: Calcium carbonate. *(Derived from general chemical knowledge as the specific compound name is not explicitly written in the provided text).*

(ix) Which of the following is the main substrate for any kind of respiration?

Answer: Glucose.

(x) Which of the following is a catabolic process?

Answer: Respiration.

(xi) The two major phases during the breakdown of glucose in respiration are (P) ..... and (Q) ..... .

Answer: P — Glycolysis, Q — Kreb's cycle.

(xii) A biology teacher asked her students to give the end products of anaerobic respiration in plants/animals. ... Who were correct?

Answer: Aayush correctly identified lactic acid as the end product of anaerobic respiration in animals. *(Note: None of the students in the options correctly provided the exact plant end products (ethanol and CO_2) together, making the book's options flawed. Based on the text, lactic acid is the correct animal product).*

✍️ B. VERY SHORT ANSWER TYPE

1. Fill in the blanks:

- **(a) ..... are the openings found on older stems.**
    - **Answer: Lenticels**
- **(b) Glycolysis occurs in the ..... of the cells.**
    - **Answer: cytoplasm**
- **(c) ..... is a respiratory substance.**
    - **Answer: Glucose**

- **(d) Rate of ..... is more than the rate of ..... in the daytime in the case of green plants.**
    - **Answer: photosynthesis, respiration**
- **(e) ..... is a chemical which absorbs oxygen of the air.**
    - **Answer: Alkaline pyrogallic acid**
- **(f) ..... is used to create vacuum (an air-free environment) to show anaerobic respiration.**
    - **Answer: Mercury** *(It displaces all air because it does not absorb CO_2 or provide O_2).*

2. Name the following:

- **(a) Energy currency of cell.**
    - **Answer:** ATP (Adenosine triphosphate).
- **(b) Oxidative breakdown of carbohydrates to release energy.**
    - **Answer:** Respiration.
- **(c) An organism which respires throughout life anaerobically.**
    - **Answer:** Yeast (or certain bacteria/fungi).
- **(d) A common phase in both aerobic and anaerobic respiration.**
    - **Answer:** Glycolysis.
- **(e) A chemical which removes CO_2 from the air.**
    - **Answer:** Soda lime (or Potassium Hydroxide / KOH).

3. Do the plants respire all day and all night or only at night?

- **Answer:** Plants respire continuously **all day and all night**.

4. Mention if the following statements are true or false. If false, rewrite them correctly.

- **(a) Aerobic respiration of one mole of glucose yields 138 ATP.**
    - **Answer: False.** Aerobic respiration of one mole of glucose yields **38 ATP**.

- **(b) Anaerobic respiration in plants yields lactic acid.**
    - **Answer: False.** Anaerobic respiration in plants yields **ethyl alcohol (ethanol) and carbon dioxide**. *(Lactic acid is produced in animals).*
- **(c) Carbon dioxide readily dissolves in lime water.**
    - **Answer: False.** Carbon dioxide turns lime water **milky** by reacting with it (it readily dissolves in Potassium Hydroxide / KOH).
- **(d) All leaves of a green plant normally respire anaerobically at night.**
    - **Answer: False.** All leaves of a green plant normally respire **aerobically** at night (they take in oxygen and give out CO_2).

📋 C. SHORT ANSWER TYPE

1. Define:

- **(a) Respiration:** A catabolic process in which living cells oxidise organic substances (glucose) to release carbon dioxide, water vapour, and energy.
- **(b) Combustion:** A fast, non-cellular, physico-chemical process occurring at high temperatures in a single step, liberating energy entirely as heat and light.
- **(c) ATP:** Adenosine Triphosphate; the chemical compound acting as the immediate "energy currency" providing energy for all activities inside a living cell.

2. Name:

- **(a) Two raw materials of respiration.**
    - **Answer:** Glucose and Oxygen.
- **(b) Two end products of aerobic respiration.**
    - **Answer:** Carbon dioxide and Water (along with ATP).
- **(c) Two microbes which normally respire anaerobically.**
    - **Answer:** Yeast and certain bacteria.

3. Write the full form of ATP and ADP.

- **Answer:**

- **ATP:** Adenosine Triphosphate.
- **ADP:** Adenosine Diphosphate.

4. What happens to the energy liberated in respiration?

- **Answer:** A large part of the energy liberated is converted into chemical energy stored in the form of ATP (used for cellular activities), and the remaining part is released as heat energy.

5. How is the tilling of the soil useful for the crops growing in it?

- **Answer:** Tilling or ploughing creates tiny air spaces around the soil particles. This provides a direct source of trapped oxygen for the general surface of the roots to absorb for aerobic respiration.

6. Why is it usually difficult to demonstrate respiration in green plants? *(From page 76)*

- **Answer:** During the daytime, green plants actively perform photosynthesis, which immediately uses up the carbon dioxide released during respiration. Therefore, to demonstrate respiration (the release of CO_2), the plant must be kept in the dark to completely stop photosynthesis.

7. Can cell respiration occur in any organism at a temperature of about 65°C? Give reason. *(From page 76)*

- **Answer: No**, cell respiration cannot occur at 65°C.
    - **Reason:** Respiration is a biochemical process entirely controlled by specific cellular enzymes. At a high temperature like 65°C, these delicate enzymes are destroyed (denatured) and lose their function, permanently stopping the respiration process.

📕 D. LONG ANSWER TYPE

1. Distinguish between the following pairs:

- **(a) Aerobic and anaerobic respiration:** Aerobic respiration requires oxygen, results in the complete breakdown of glucose, and yields a large amount of energy (38 ATP) along with CO_2 and water. Anaerobic

respiration occurs without oxygen, results in incomplete breakdown, and yields very little energy (2 ATP) along with ethanol and CO_2.

- **(b) Respiration and combustion:** Respiration is a slow, cellular, enzyme-controlled biochemical process that occurs at body temperature and traps energy as ATP. Combustion is a fast, non-cellular, physico-chemical process requiring high ignition heat that releases all energy as heat and light in a single step.
- **(c) Photosynthesis and respiration:** Photosynthesis is an anabolic (constructive) process that occurs only in chlorophyll-containing cells during light, consuming CO_2 and water to manufacture glucose and release O_2, resulting in weight gain. Respiration is a catabolic (destructive) process occurring in all cells at all times, breaking down glucose using O_2 to release CO_2 and water, resulting in weight loss.
- **(d) Stomata and lenticels:** Stomata are tiny microscopic pores primarily found on the surface of leaves used for gaseous exchange. Lenticels are permanent openings found on the bark of older, mature stems that serve as inlets for oxygen.

2. How do the following structures help in respiration in plants?

- **(a) Lenticels:** They act as permanent openings on mature stems, allowing atmospheric oxygen to diffuse into the inner living tissues of the stem.
- **(b) Stomata:** They open to allow the direct diffusion of oxygen from the air into the leaf cells and the exit of carbon dioxide.
- **(c) Root hairs:** They are highly permeable and absorb the oxygen trapped in the microscopic air spaces between soil particles.

3. How are aerobic and anaerobic respirations different in plants?

- **Answer:**
    1. **Oxygen:** Aerobic uses oxygen; Anaerobic proceeds without oxygen.
    2. **Breakdown:** Aerobic is complete oxidation of glucose; Anaerobic is incomplete breakdown.
    3. **End Products:** Aerobic produces Carbon dioxide and water; Anaerobic produces Ethyl alcohol and Carbon dioxide.

4. **Energy:** Aerobic yields 38 ATP per glucose molecule; Anaerobic yields only 2 ATP.
5. **Duration:** Aerobic occurs normally throughout the plant's life; Anaerobic occurs only temporarily during unfavorable conditions (like water-logging).

4. Explain why respiration is said to be the reverse of photosynthesis. (HOTS)

- **Answer:** They are exact biological opposites because the **raw materials of one are the end products of the other**. Photosynthesis takes in CO_2 and H_2O and uses light energy to build Glucose and release O_2 (Anabolic). Respiration takes that Glucose and O_2, breaks it down to release energy, and gives out CO_2 and H_2O as waste (Catabolic).

5. What is respiration? How are respiration and burning similar and how are they different?

- **Definition:** Respiration is a catabolic process by which living cells oxidise organic substances (glucose) to release carbon dioxide, water vapour, and energy.
- **Similarities:** Both processes liberate energy, and both involve the breakdown of carbon-containing substances to produce Carbon Dioxide (CO_2) and Water (H_2O) as end products.
- **Differences:** Respiration is a cellular, multi-step, enzyme-driven process producing ATP without light. Burning is a non-cellular, single-step process driven by high heat that produces fire (light and heat).

6. Describe one experiment each you would perform to demonstrate the following phenomena:

- **(a) The germinating seeds produce heat:** Take two thermoflasks. Place live germinating seeds in Flask A and boiled seeds (washed with formalin to prevent decay) in Flask B. Insert a thermometer into the cotton plug of both. After a few hours, the thermometer in Flask A will show a higher reading, proving respiration produces heat.
- **(b) The germinating seeds give out carbon dioxide:** Place germinating seeds in Flask A and boiled seeds (with antiseptic) in Flask B. Seal them

tightly. After a few days, remove the corks and tilt the flasks over a test tube containing limewater. The heavy gas from Flask A flows down and turns the limewater milky, proving CO_2 was given out.

- **(c) The germinating seeds can respire even in total absence of air:** Peel the seed coats off 8-10 soaked pea seeds and push them into a test tube completely filled and inverted in a beaker of mercury (creating an oxygen-free environment). After two days, the mercury level falls as gas collects at the top. Introducing a KOH stick makes the mercury rise again (as KOH absorbs CO_2), proving seeds respired anaerobically.

7. Draw a neat and labelled diagram showing the experimental set up (initial and final stages) to demonstrate anaerobic respiration in germinating seeds.

- *(Note: Since I cannot draw on your paper, here is how you must sketch it for full marks!)* 🖍️
    - **Initial Stage:** Draw a beaker filled with mercury. Draw a test tube completely filled with mercury inverted into the beaker. Draw peeled seeds floating at the very top of the test tube.
    - **Final Stage (After 2-3 days):** Draw the exact same setup, but the mercury level in the test tube has dropped significantly. Label the empty space at the top as "CO_2 Gas". Draw a KOH stick being inserted from the bottom.

🟢 E. STRUCTURED/APPLICATION/SKILL TYPE

1. The following two chemical reactions are supposed to indicate a certain process occurring in the green plants under two different conditions:

(a) $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + \ldots\ldots + 38ATP$

(b) $C_6H_{12}O_6 \rightarrow \ldots\ldots\ldots\ldots + 2CO_2 + 2ATP$

- **(i) Fill in the blanks of each reaction.**
    - **(a)** $6H_2O$ (Water)
    - **(b)** $2C_2H_5OH$ (Ethyl alcohol / Ethanol)
- **(ii) Name the process represented by the two chemical equations.**
    - **(a)** Aerobic respiration.

- **(b)** Anaerobic respiration.

- **(iii) What are the conditions under which the two reactions (a) and (b) are occurring?**
  - Reaction (a) occurs in the presence of free oxygen.
  - Reaction (b) occurs in the complete absence of oxygen.

2. The experimental set-up given below demonstrates anaerobic respiration in germinating seeds. Read the information below the figure and answer the following questions:

- **(a) Why do we fill mercury in the test tube? Can we do the same experiment in water or air?**
  - Mercury is used to create an oxygen-free (anaerobic) vacuum seal because it does not absorb carbon dioxide and does not provide dissolved oxygen. We **cannot** use water because seeds might extract dissolved oxygen from it, and water absorbs the released CO_2. We **cannot** use air because it contains oxygen, which would trigger aerobic respiration instead.

- **(b) Name the gas released by the germinating seeds, shown (A) in the figure.**
  - Carbon dioxide (CO_2).

- **(c) Can we do the above experiment with the seeds without peeling them?**
  - Yes, the experiment will still work, but peeling off the seed coat (testa) removes a physical barrier, which heavily *hastens* (speeds up) the outward diffusion of the CO_2 gas.

- **(d) How can we test the gas released in the above set-up? Explain in short.**
  - Introduce a stick of Potassium Hydroxide (KOH) into the test tube from the bottom. It will float up through the mercury into the gas space and absorb the carbon dioxide. Consequently, the level of mercury will immediately rise again, proving the gas was CO_2.

- **(e) How can we set the 'control set-up' for the above experiment?**
    - Take an identical set-up but replace the live germinating seeds with an equal number of **boiled (dead) seeds**. These boiled seeds must be washed with an antiseptic (like carbolic acid) to prevent any bacterial decay from generating false gas.

3. In order to study and prove a particular physiological process in plants, the following experiment was set up. Study the same and then answer the questions that follow:

- **(a) Name the physiological process being studied.**
    - Respiration (Specifically, proving that CO_2 is produced during aerobic respiration).
- **(b) What is the function of soda lime in the bottle 'A' and why is limewater placed in bottle 'B'?**
    - **Soda lime in 'A':** Absorbs any carbon dioxide naturally present in the incoming atmospheric air.
    - **Limewater in 'B':** Acts as an indicator to confirm that the air entering flask C is completely CO_2-free (the limewater should remain clear and not turn milky).
- **(c) What change would you expect to observe in bottle 'D'?**
    - The limewater in bottle 'D' will turn milky.
- **(d) Represent the physiological process named in (a) above in the form of a chemical equation.**
    - $C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{enzyme}} 6CO_2 + 6H_2O + 38ATP$
- **(e) In order to obtain accurate results, the bottle 'C' should be covered with a piece of black cloth. Why?**
    - To completely prevent photosynthesis. If exposed to light, any green parts of the germinating seeds might perform photosynthesis and use up the CO_2 just liberated by respiration, which would ruin the result in bottle 'D'.

- **(f) If bottle 'C' was fitted with a 3-holed rubber stopper and a thermometer were introduced in such a way that its bulb reaches close to the germinating seeds, what would you observe? Why?**
    - You would observe a **rise in temperature** on the thermometer. This happens because respiration is a catabolic process that releases energy, a significant portion of which is liberated as heat.

4. The given diagram refers to an apparatus which is used to demonstrate a physiological process:

- **(a) What is the purpose of keeping potassium hydroxide solution in the test tubes X and Y?**
    - Potassium hydroxide (KOH) acts as a strong absorbent for carbon dioxide. It absorbs any CO_2 released by the respiring seeds.
- **(b) Why has the coloured water risen in tubing 1?**
    - The live germinating seeds in Test-Tube X consume oxygen for respiration and release CO_2. The KOH immediately absorbs this released CO_2. This creates a partial vacuum (a drop in air pressure) inside Test-Tube X, causing the outside atmospheric pressure to push the colored water up Tubing 1 to fill the space.
- **(c) What is the purpose of keeping boiled peas soaked in a disinfectant in test tube Y?**
    - This acts as the **control**. Boiling kills the seeds so they cannot respire. The disinfectant prevents bacterial or fungal growth, which would otherwise respire, consume oxygen, and ruin the control setup.
- **(d) Name the biological process which causes the above rise.**
    - Aerobic respiration.
- **(e) Define the biological process shown in the experiment.**
    - Respiration is a catabolic process by which living cells oxidise organic substances (glucose) to release carbon dioxide, water vapour, and energy.

5. Given alongside is a set of six experimental set-ups (A-F), kept in this state for about 24 hours.

- **(a) In how many flasks, the different plant parts have been kept under observation?**
    - In **five** flasks (Flasks A, B, C, D, and E).
- **(b) What is the purpose of keeping a test-tube containing limewater in each flask?**
    - Limewater is used as an indicator to test for the production of carbon dioxide by the enclosed plant parts (it turns milky in the presence of CO_2).
- **(c) In which tube/tubes the limewater will turn milky?**
    - The limewater will turn milky in tubes **A, B, C, D, and E**.
- **(d) What is the purpose of the set-up F?**
    - Set-up F acts as the **control**. Because it contains no plant parts, its limewater will not turn milky, proving that the CO_2 in the other flasks was generated specifically by the living plant parts and not by the flask environment itself.
- **(e) What conclusion can you draw from this experiment?**
    - The conclusion is that **every living part of a plant** (flowers, leaves, germinating seeds, roots, and stem pieces) actively respires and produces carbon dioxide.